

EEE 120

Lab 2 Answer Sheet

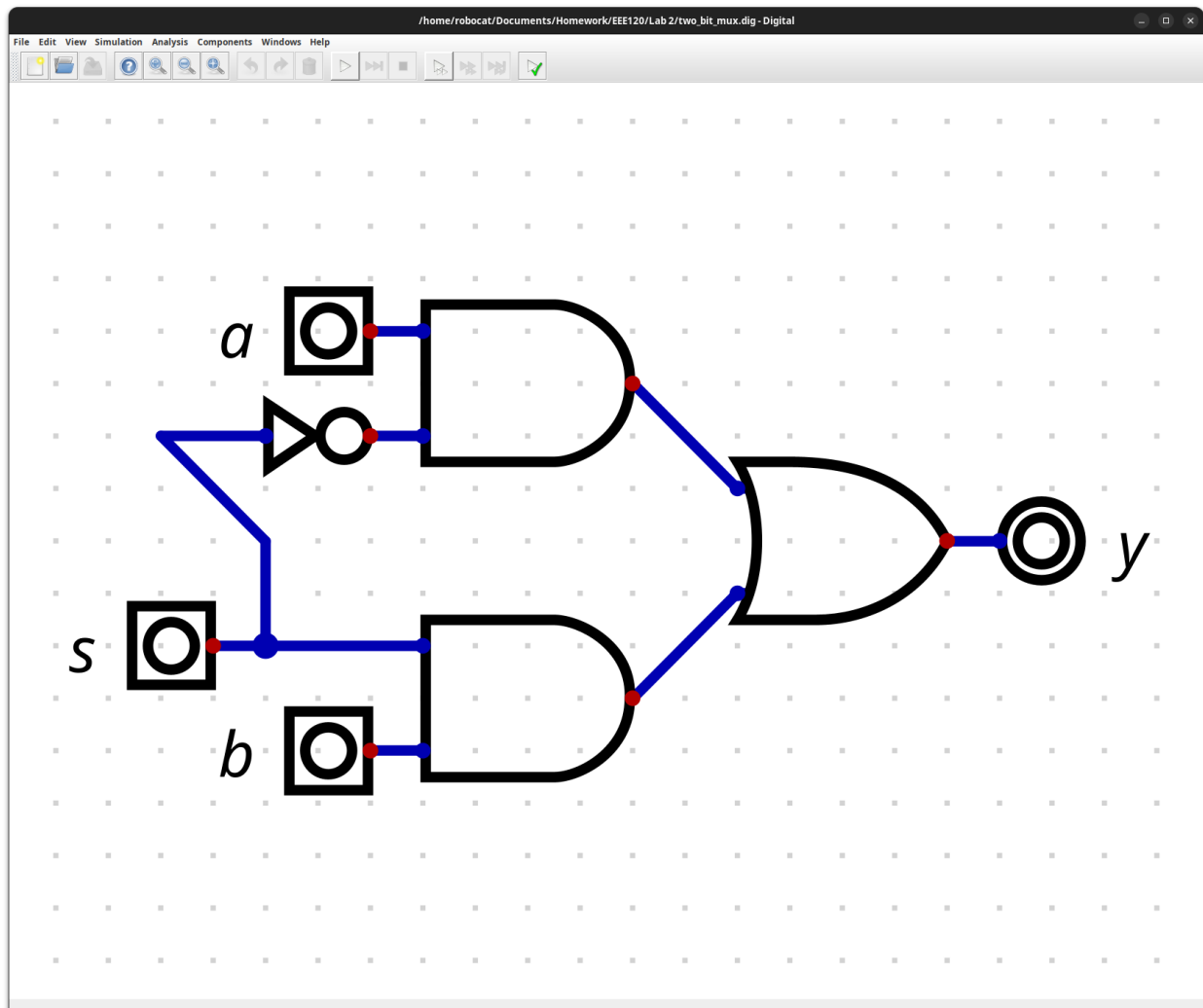
Multiplexers, Decoders and the Arithmetic and Logic Unit (ALU)

Name: Dylan Shah

Instructor/Time: Dr. Steve Millman Tu Th 3:00 PM - 4:15 PM Date: January 13, 2026

Task 2-1: Build and Test a 1-Bit 2:1 Multiplexer

Include a picture of your Digital circuit here:

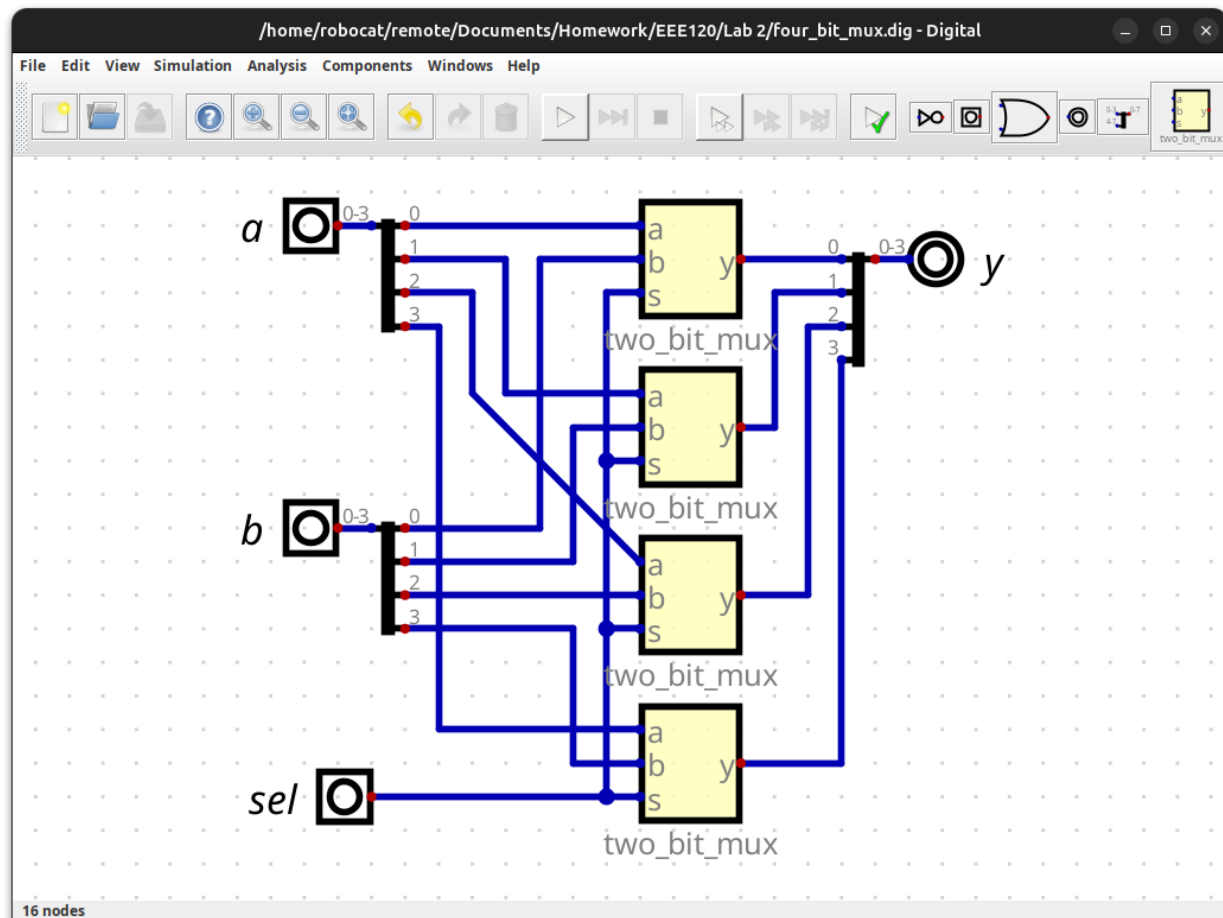


Please comment on the single biggest issue you were facing when designing the circuit.

I did not have any issues designing the circuit.

Task 2-2: Build a 4-Bit 2:1 Multiplexer

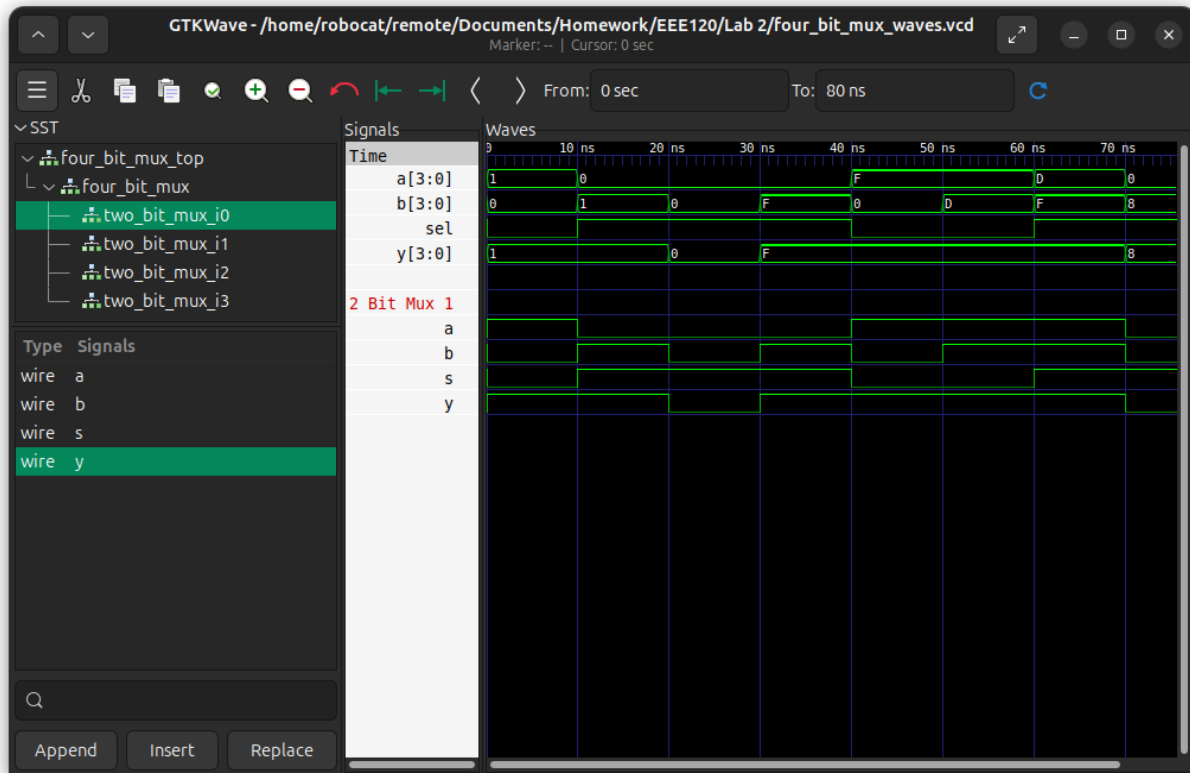
Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

The connection between a.2 and the mux would have the wire going where I already routed wires for b, so I had to make a diagonal wire for it, unfortunately.

Include a picture of your simulation (timing diagram) here:



Which tests did you perform and why? Use the following table to describe your test sequence. You need to make sure to perform enough tests to check the circuit for possible incorrect behavior.

Test Stimulus	Test Motivation	Pass/Fail
1_0_1_0	Test the first bit is switched when sel is off.	PASS
1_1_0_1	Test the first bit is switched when sel is on.	PASS
0_1_0_0	All zeros pass through	PASS
F_1_0_F	All ones pass though	PASS
F_0_F_0	All ones pass though	PASS
F_0_F_D	Changing inactive input has no effect	PASS

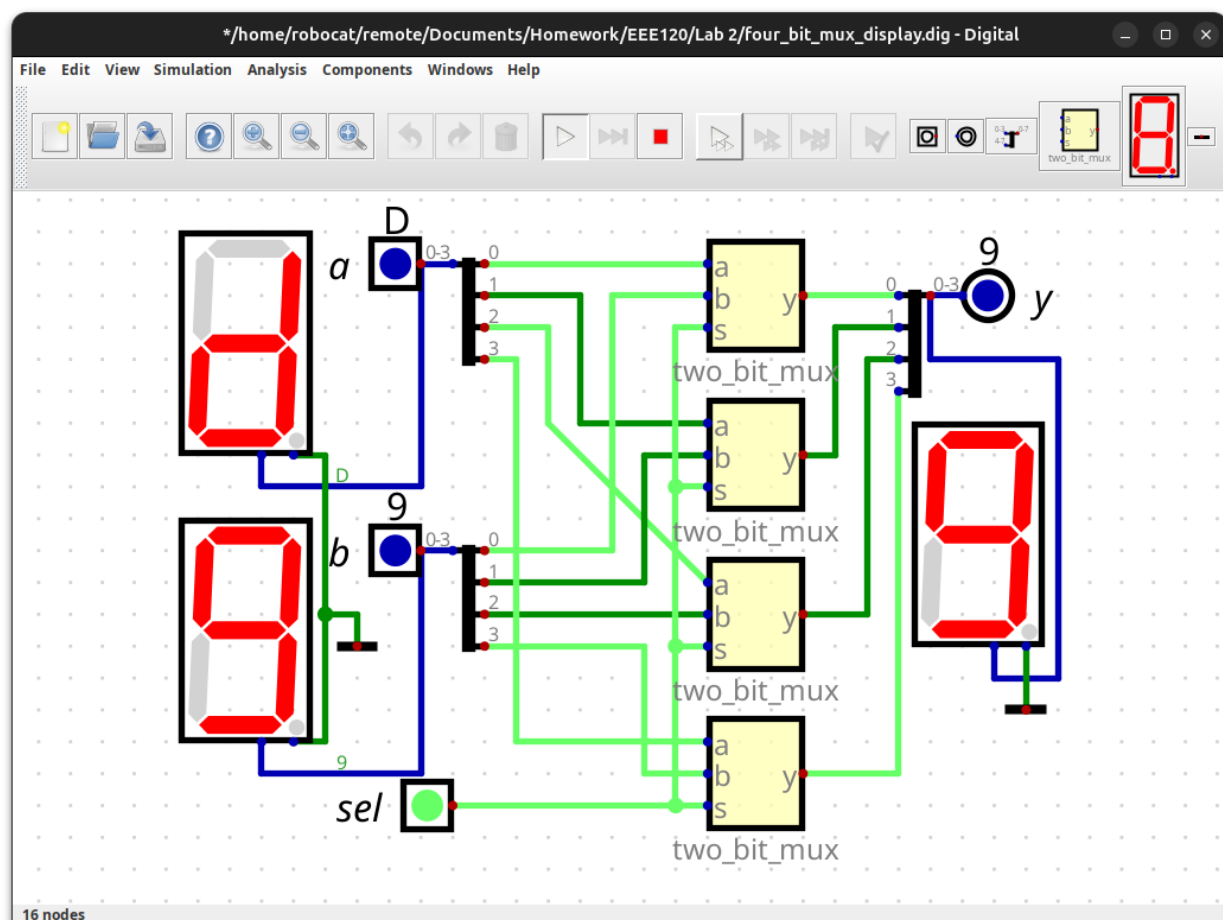
F_1_D_F	Changing inactive input has no effect	PASS
8_1_0_8	Order of bits is same	PASS

Please comment on the single biggest issue you were facing when simulating the circuit.

I did not face any issues simulating the circuit.

Task 2-3: Add 7-Segment Displays to Your Circuit

Include a picture of your Digital 7-Segment Display here:

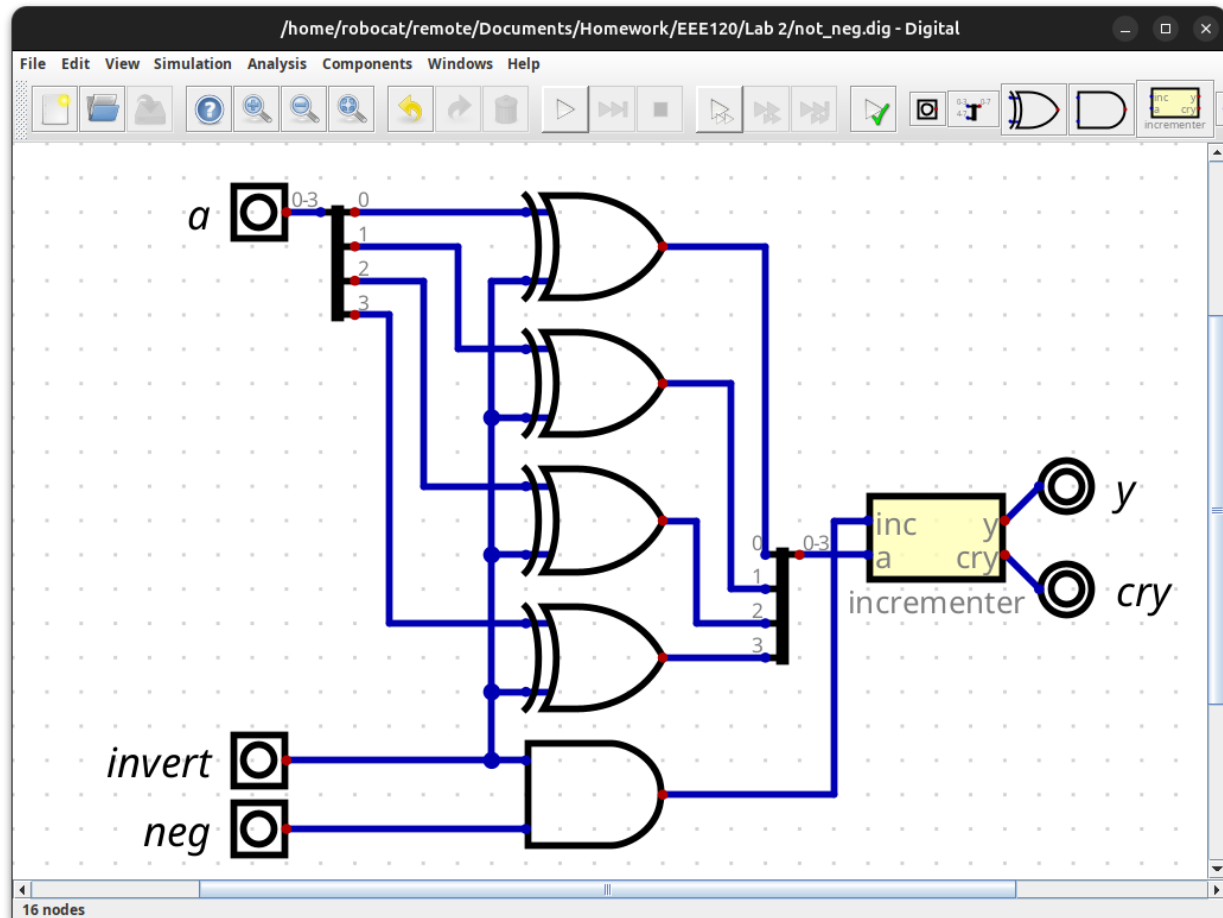


Please comment on the single biggest issue you were facing when adding the displays.

The component was much larger than I thought, and I struggled to find space for it.

Task 2-4: Build the NOT/NEG Circuit

Include a picture of your Digital circuit here:

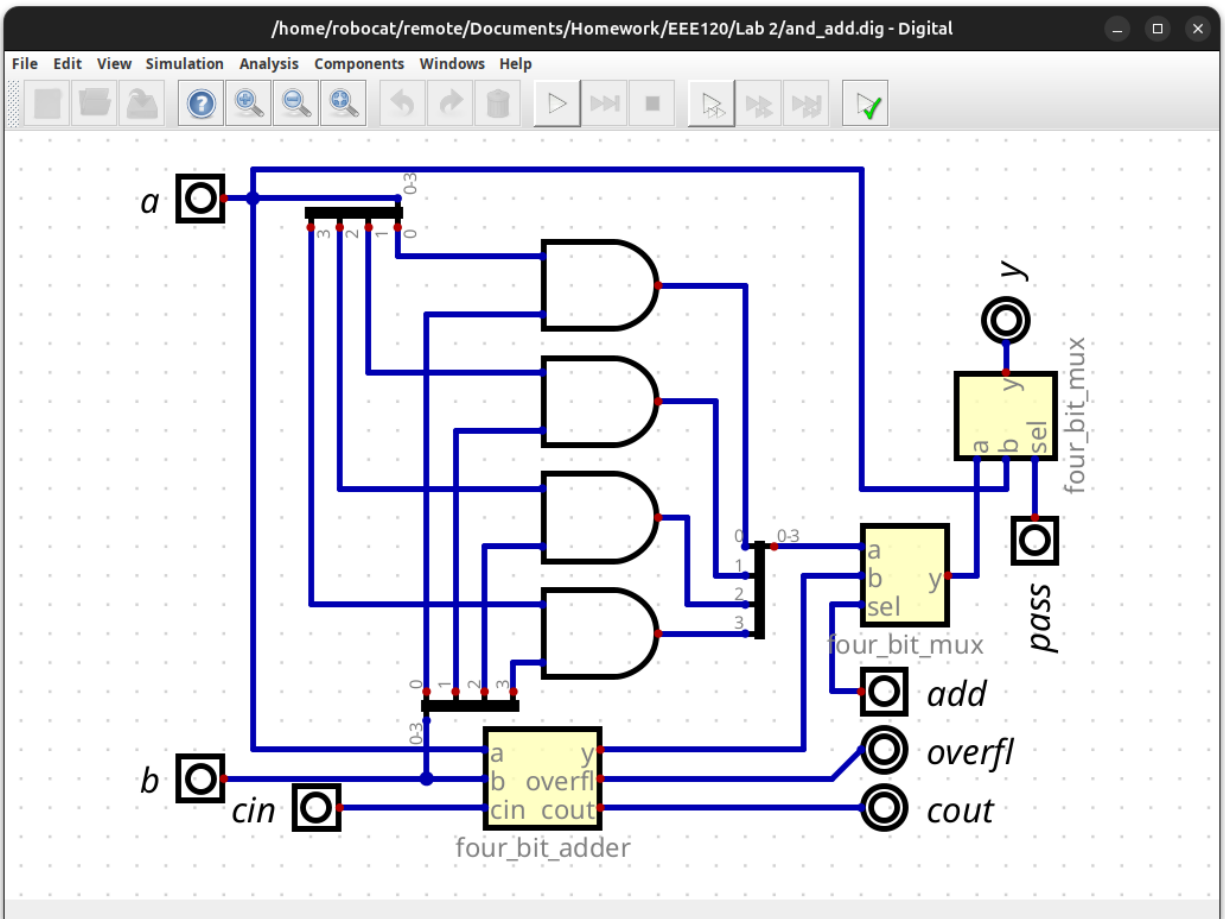


Please comment on the single biggest issue you were facing when designing the circuit.

The labels on my incrimenter were in the opposite order of what was in the example, which tripped me up on my first attempt routing the wires.

Task 2-5: Build the AND/ADD Circuit

Include a picture of your Digital circuit here:

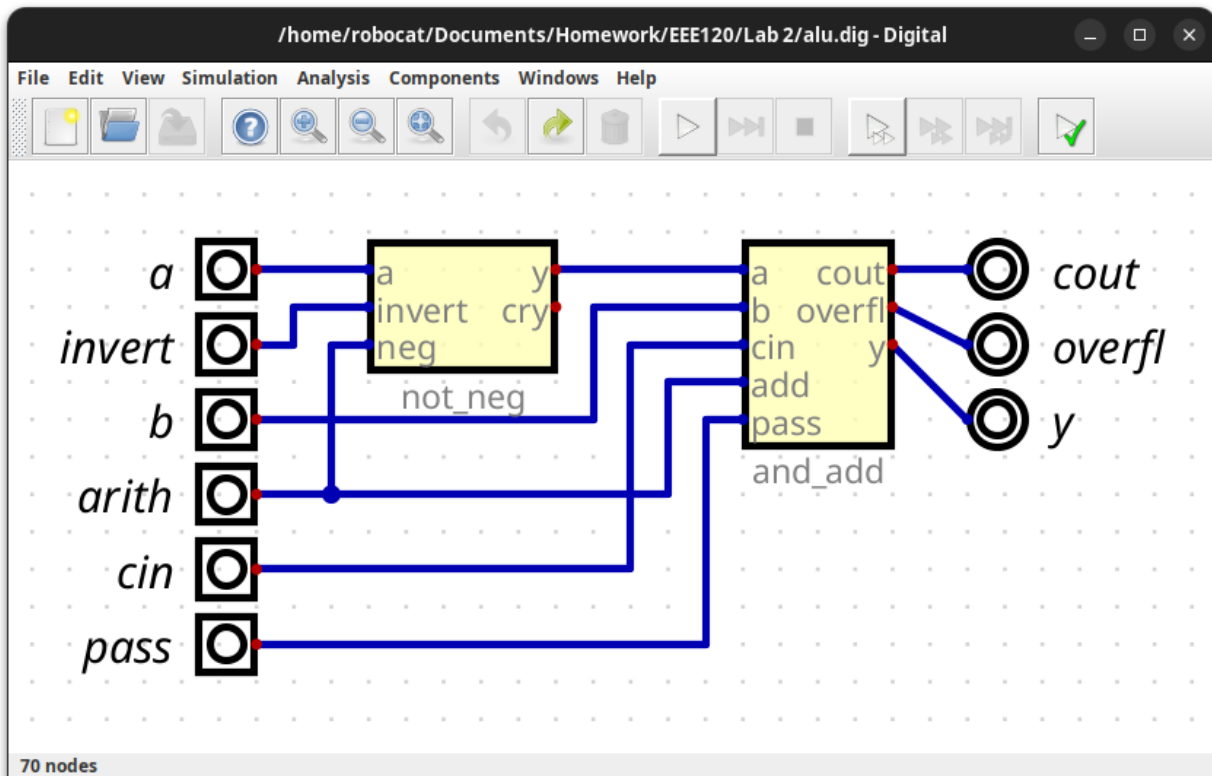


Please comment on the single biggest issue you were facing when designing the circuit.

I did not experience any issues creating this circuit.

Task 2-6: Build and Test the ALU Circuit

Include a picture of your Digital circuit here:



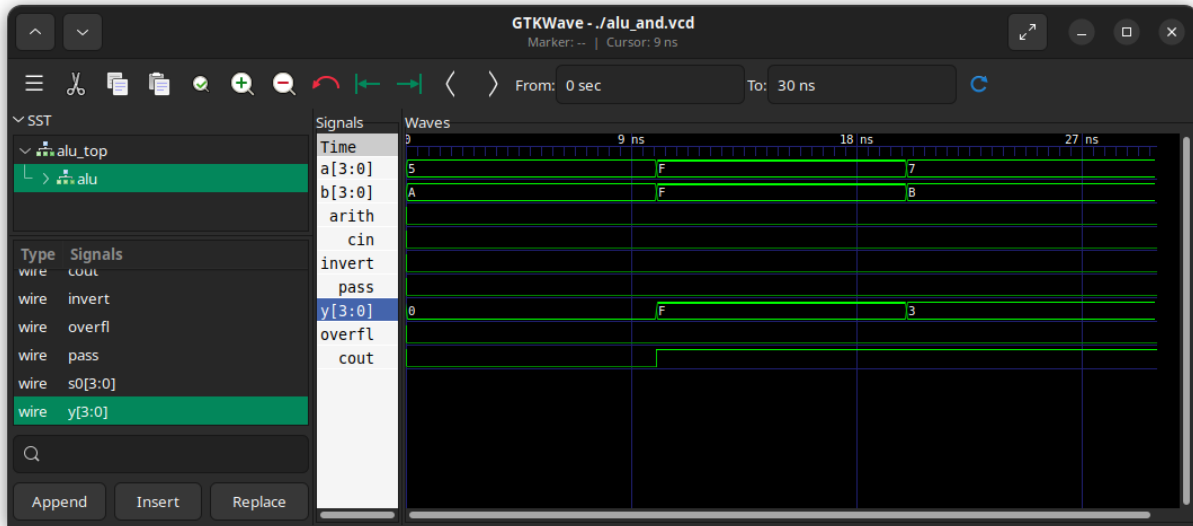
Please comment on the single biggest issue you were facing when designing the circuit.

I did not have any difficulties designing the circuit.

Please complete the ALU function definition table shown below:

arith	invert	pass	Function
0	0	0	AND, AB
0	0	1	PASS THROUGH, A
0	1	0	NOT A AND B, $\bar{A}B$
0	1	1	INVERT A, $\bar{A}$
1	0	0	Arithmetic Sum, A+B
1	0	1	PASS THROUGH, A
1	1	0	SUBTRACT, B-A
1	1	1	NEGATE A, -A

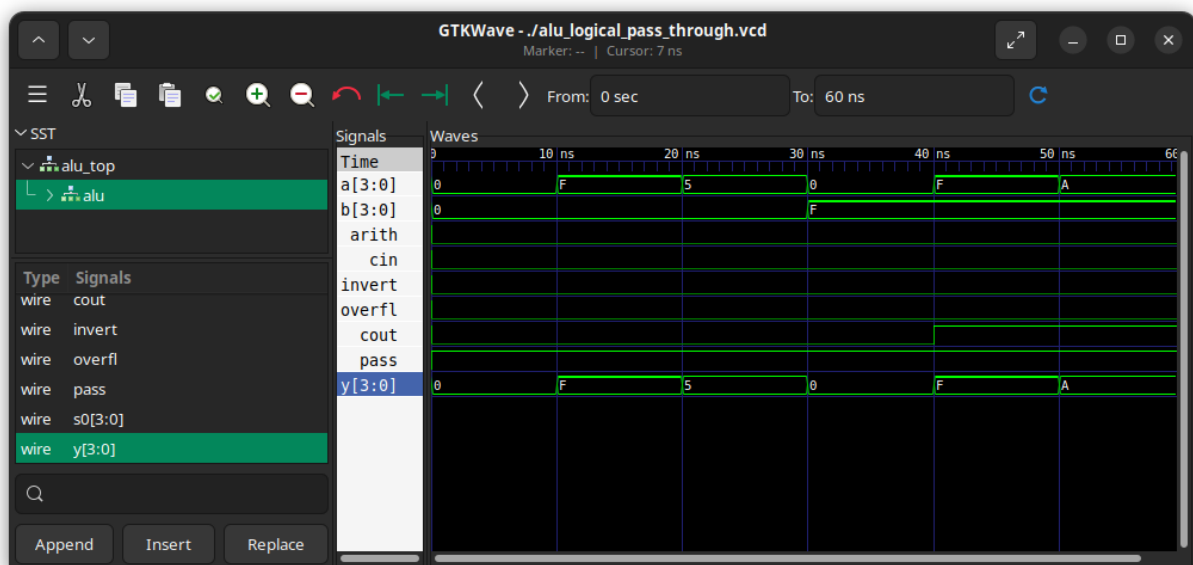
Include all pictures of simulations (timing diagrams) for each function here. With each picture, paste the test_vals from your stimulus file for that test.



test_vals[0] = 18'h0_0_0_5_A; // Ensure all bits can turn off

test_vals[1] = 18'h0_F_0_F_F; // Ensure all bits can turn on

test_vals[2] = 18'h0_3_0_7_B; // Ensure some bits can turn on and some off



test_vals[0] = 18'h0_0_1_0_0; // Ensure there are no stuck on bits

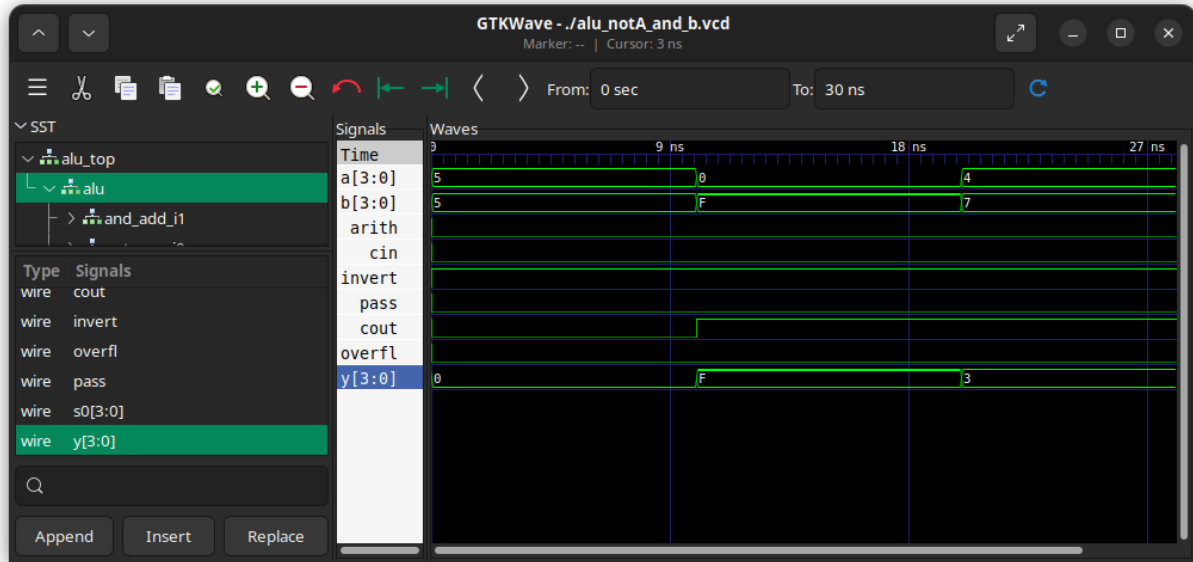
test_vals[1] = 18'h0_F_1_F_0; // Ensure there are no stuck off bits

test_vals[2] = 18'h0_5_1_5_0; // Ensure bit ordering stays the same

test_vals[3] = 18'h0_0_1_0_F; // Ensure B does not affect A

test_vals[4] = 18'h0_F_1_F_F; // Ensure B does not affect A

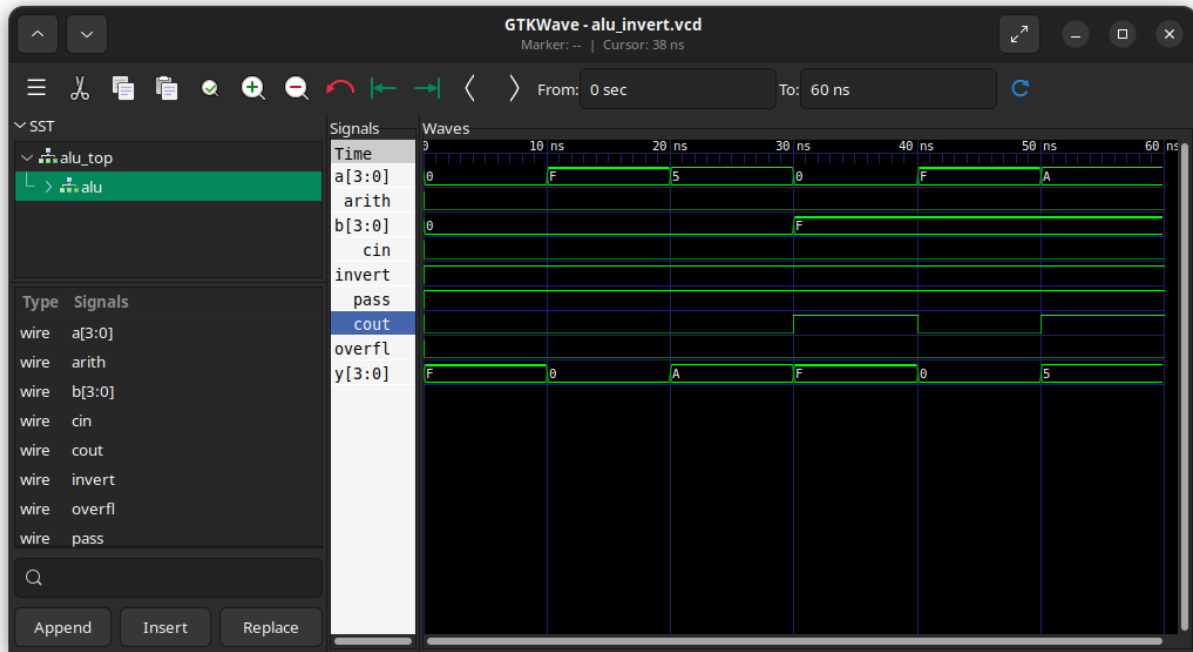
test_vals[5] = 18'h0_A_1_A_F; // Ensure B does not affect A



test_vals[0] = 18'h0_0_2_5_5; // Ensure all bits can turn off

test_vals[1] = 18'h0_F_2_0_F; // Ensure all bits can turn on

test_vals[2] = 18'h0_3_2_4_7; // Ensure some bits can turn on and some off0



test_vals[0] = 18'h0_F_3_0_0; // Ensure there are no stuck on bits

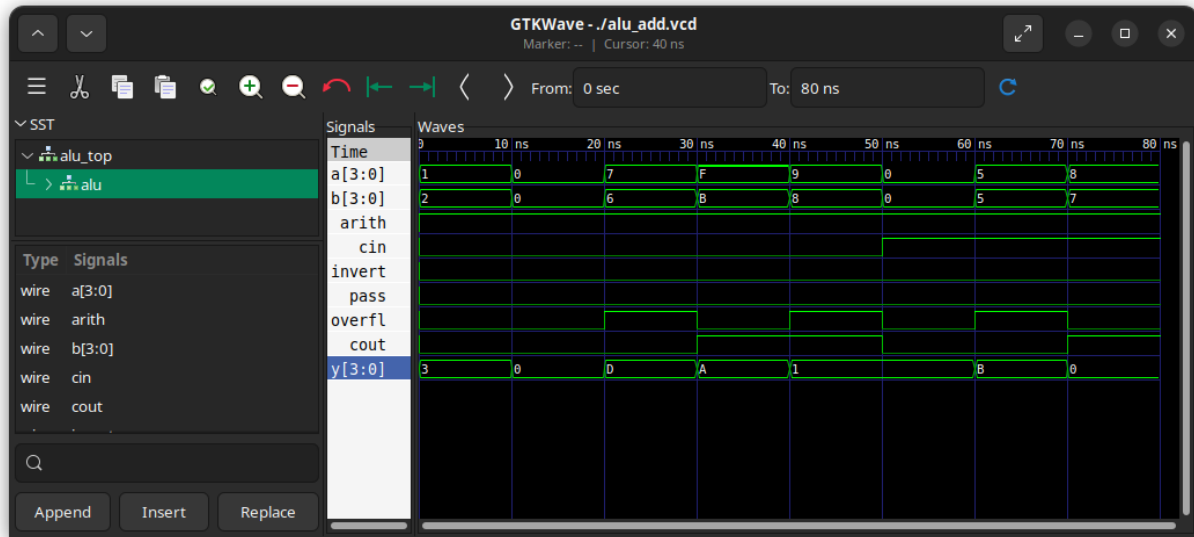
test_vals[1] = 18'h0_0_3_F_0; // Ensure there are no stuck off bits

test_vals[2] = 18'h0_A_3_5_0; // Ensure bit ordering stays the same

test_vals[3] = 18'h0_F_3_0_F; // Ensure B does not affect A

test_vals[4] = 18'h0_0_3_F_F; // Ensure B does not affect A

test_vals[5] = 18'h0_5_3_A_F; // Ensure B does not affect A



```
test_vals[0] = 18'h0_3_4_1_2; // Check small numbers add correctly
```

```
test_vals[1] = 18'h0_0_4_0_0; // Check that there are no stuck bits
```

```
test_vals[2] = 18'h2_D_4_7_6; // Check that one large unsigned number and a smaller
number add correctly
```

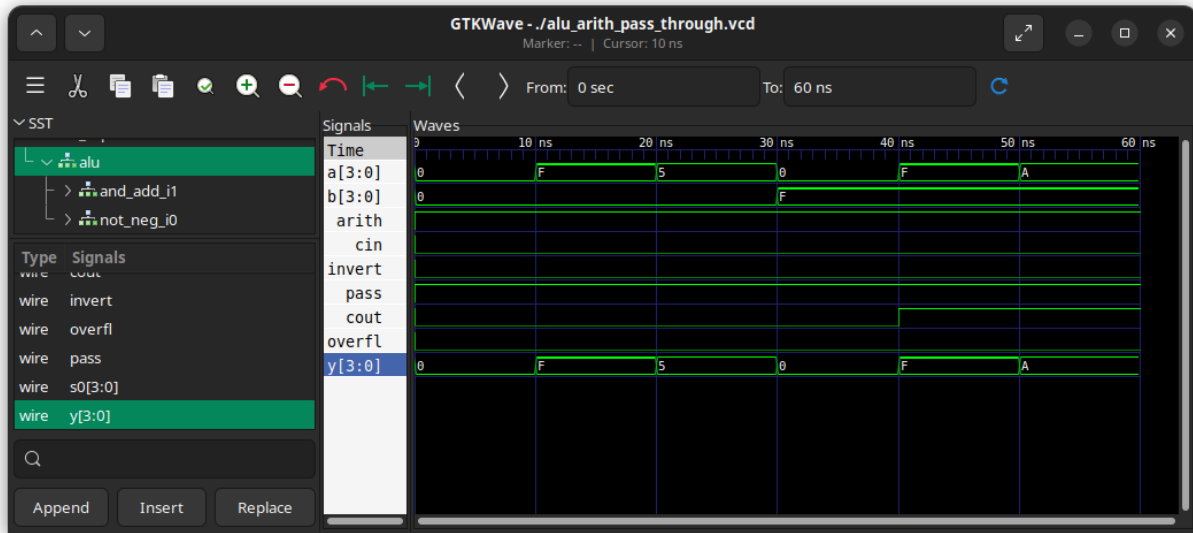
```
test_vals[3] = 18'h1_A_4_F_B; // Check that two negative numbers add correctly with no
overflow
```

```
test_vals[4] = 18'h3_1_4_9_8; // Check that overflow happens correctly.
```

```
test_vals[5] = 18'h0_1_C_0_0; // Check Cin adds correctly
```

```
test_vals[6] = 18'h2_B_C_5_5; // Check Cin adds correctly
```

```
test_vals[7] = 18'h1_0_C_8_7; // Check Cin adds correctly, and  $-(-8) = -8$ , as there is no
positive representation
```



`test_vals[0] = 18'h0_0_5_0_0; // Ensure there are no stuck on bits`

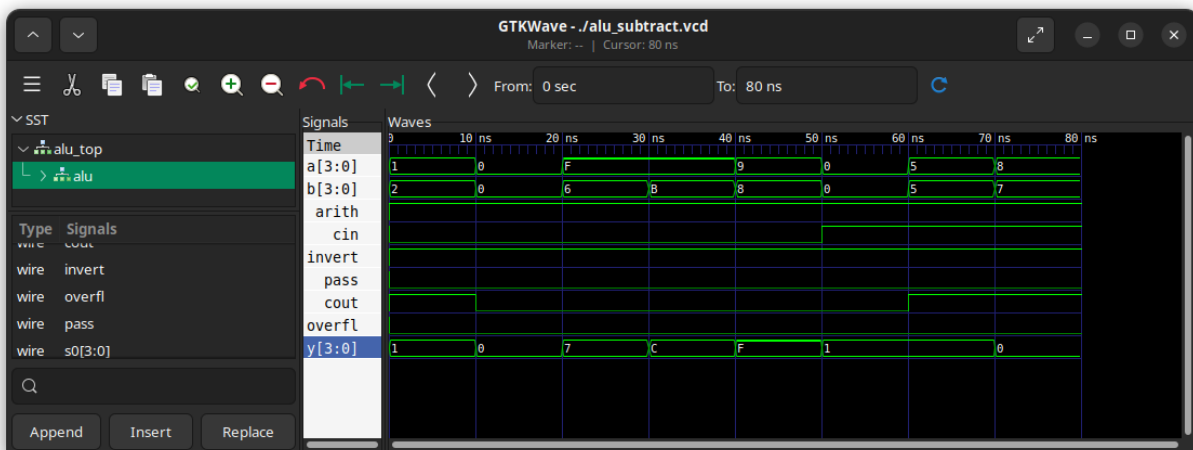
`test_vals[1] = 18'h0_F_5_F_0; // Ensure there are no stuck off bits`

`test_vals[2] = 18'h0_5_5_5_0; // Ensure bit ordering stays the same`

`test_vals[3] = 18'h0_0_5_0_F; // Ensure B does not affect A`

`test_vals[4] = 18'h0_F_5_F_F; // Ensure B does not affect A`

`test_vals[5] = 18'h0_A_5_A_F; // Ensure B does not affect A`



`test_vals[0] = 18'h1_1_6_1_2; // Check small numbers subtract correctly`

`test_vals[1] = 18'h0_0_6_0_0; // Check that there are no stuck bits`

`test_vals[2] = 18'h0_7_6_F_6; // Check that a number minus a negative number is their sum`

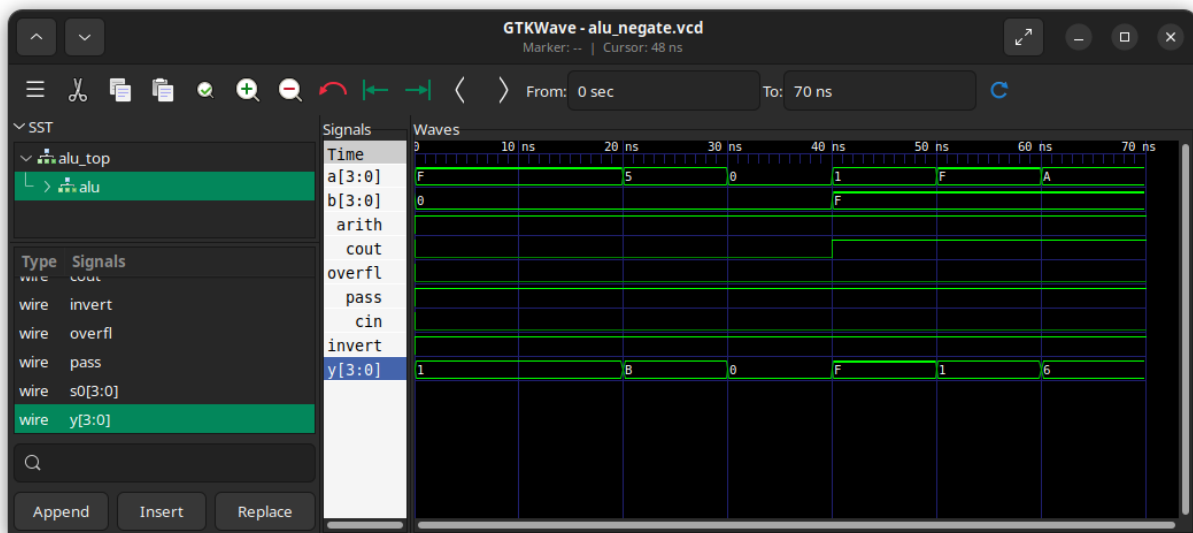
`test_vals[3] = 18'h0_C_6_F_B; // Check that two negative numbers subtract correctly with no overflow`

`test_vals[4] = 18'h0_F_6_9_8; // Check that overflow happens correctly.`

`test_vals[5] = 18'h0_1_E_0_0; // Check Cin adds to the difference correctly`

`test_vals[6] = 18'h1_1_E_5_5; // Check Cin adds to the difference correctly`

`test_vals[7] = 18'h1_0_E_8_7; // Check Cin adds to the difference correctly, and  $-(-8) = -8$ , as there is no positive representation`



`test_vals[0] = 18'h0_1_7_F_0; // Ensure there are no stuck on bits`

`test_vals[1] = 18'h0_1_7_F_0; // Ensure there are no stuck off bits`

`test_vals[2] = 18'h0_B_7_5_0; // Ensure bit ordering stays the same`

`test_vals[3] = 18'h0_0_7_0_0; // Ensure  $-0 = 0$`

`test_vals[4] = 18'h0_F_7_1_F; // Ensure B does not affect A`

`test_vals[5] = 18'h0_1_7_F_F; // Ensure B does not affect A`

`test_vals[6] = 18'h0_6_7_A_F; // Ensure B does not affect A`

Did the circuit behave as expected? If no, what was wrong? Yes, the circuit works as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

I was struggling to think of what tests are needed to ensure the circuit works as expected.

Task 2-7: Create a video and submit your report

Record a short video showing your schematics in Digital and your waveforms in GTKWave. Be sure to show yourself in the video and show your screen. Explain how your circuit works – you need to convince the grader you did the lab and understand it! **Copy and paste the link to your video below. Make sure the link is working and pointing to the correct video. Remember to include the password if required. Do NOT upload your video to Canvas. It is recommended that you use Zoom to record to the cloud, pasting the link and password below.** If your circuit is not working as expected, explain in the video how it is not working and why you think it is not working.

Video Link: https://drive.google.com/file/d/1irdxSqPzMckJTAj3BQAshP16h56tdms1/view?usp=drive_link

At the beginning of your recording, say your name and the lab name. Be brief in your recording. Submit the completed template to Canvas.

Make sure all your files are in the Lab2 directory. Create a zip file of the Lab2 directory. Remember to turn in the zip file and your completed template on Canvas!

Do not include the video in the zip file! This makes the file very large and you run the risk of the zip file not uploading or taking so long to upload that your submission will be late. Remember that the submission is dated at the time the upload completes, not when it starts!

LAB 2: LAB REPORT GRADE SHEET

Name _____

**NOTE: You submit the zip file in order to show your work.
If the zip file is not submitted you will receive a 0 for this lab!**

Instructor Assessment

Grading Criteria	Max Points	Points Lost
Description of Assigned Tasks. Work Performed &		
Task 2-1: Build and Test a 1-bit 2:1 Multiplexer	5	
Task 2-2: Build a 4-Bit 2:1 Multiplexer	5	
Task 2-3: Add 7-Segment Displays to Your Circuit	5	
Task 2-4: Build the NOT/NEG Circuit	5	
Task 2-5: Build the AND/ADD Circuit	10	
Task 2-6: Build and Test the ALU Circuit	10	
Task 2-7: Create a video and submit your report.	10	
Lab Score (50 points total)	Points	
	Late Lab	
	Lab Score	